Reader Report: Week 10

Bayesian Optimization and Adaptive Experimental Design

Daniel Hardesty Lewis (dl3645)

Spring 2026

Summary

Bayesian Optimization (BO), as outlined in Garnett's *Bayesian Optimization*, formalizes the search for global optima of expensive, black-box functions through adaptive experimental design. The framework consists of two core components: a probabilistic surrogate model (typically a Gaussian Process) that maintains a belief state over the objective function, and an acquisition function (e.g., Expected Improvement) that navigates the explore-exploit trade-off to dictate the next query point. BO is an adaptive design strategy for selecting queries x_t to efficiently learn about an expensive objective. It becomes a tool for causal optimization only when the query corresponds to a well-defined intervention and the optimized quantity is an interventional estimand (e.g., $\mathbb{E}[Y \mid do(A = a)]$). Its central premise is expensive evaluation and the need to trade off exploration and exploitation via a probabilistic surrogate, rather than relying on differentiable structural equations.

Critique

The theoretical leap from synthetic controls (SC) and invariant causal prediction (ICP) to Bayesian Optimization bridges the gap between evaluating historical policy interventions and dynamically prescribing new ones. As established by **?**, SC weights and causal identifiability rely on fine-grained individual-level potential outcomes, independent causal mechanisms, and structural conditions like stability, overlap, and similarity across selected donors (Theorem 1). In the context of urban modeling, scaling from individual resident opinions to neighborhood-level protest events suggests that the linear structure in SC arises from expectation and aggregation, not necessarily linear individual mechanisms. This ensures that identifiability is treated as a rigorous property of the target and donor composition, not just a computational convenience.

Furthermore, applying BO to social domains introduces critical tension. In domain generalization, the environment $e \in \mathcal{E}$ indexes a distribution or regime generating samples; it is not typically an active choice variable or constraint. If the dynamic data generating process experiences structural mechanism shifts (as opposed to mere covariate shifts under a stable causal mechanism), the stationary covariance structure of the GP surrogate becomes miscalibrated. Incorporating causal structure directly into the surrogate model (such as using Causal Bayesian Optimization (CBO) (**?**), which explicitly leverages causal graphs and *do*-calculus to balance the observation and intervention tradeoff) is necessary for deploying adaptive experimental design across unstable temporal or geographic regimes.

Questions

1. When extending BO to causal domains, confounding (the mismatch between $\mathbb{E}[Y \mid X = x]$ and $\mathbb{E}[Y \mid do(X = x)]$ due to unmeasured common causes) cannot be resolved merely by

decomposing epistemic variance and aleatoric observation noise in a GP posterior. How can dynamic BO models integrate identification assumptions (e.g., back-door adjustments or instrumental variables) before utilizing posterior variance for exploration?

2. If the underlying structural causal model shifts dynamically over time (e.g., a policy intervention alters the true mechanisms rather than just inducing covariate shift), how do formal nonstationary frameworks (like time-varying GP bandits (**?**) or Dynamic Causal Bayesian Optimization (**?**)) track the drifting optimum without manually re-initializing the GP prior?

Project Update: Zoning Policy as Adaptive Experimental Design

Abstract

My project formalizes the prediction of formal protest petitions against housing developments as an Empirical Risk Minimization (ERM) problem subject to spatial and temporal domain shifts. Building on our invariant causal prediction framing, this update explores framing municipal rezoning interventions not as static longitudinal treatments, but as an ongoing *constrained Bayesian Optimization* problem managed by the city council.

The Questions (What, Why, How)

What: Can we model municipal zoning pipeline approvals as a structured Bayesian Optimization process where planners adaptively test spatial policy interventions corresponding to well-defined potential outcomes under an unknown NIMBY-resistance constraint?

Why: Standard retrospective analyses (like Difference-in-Differences) evaluate spatial shocks statically. However, cities operate sequentially: granting density bonuses, observing noisy protests, and calibrating future approvals. By defining the city's behavior through BO constraint satisfaction algorithms (e.g., SAFEOPT) guaranteeing safe exploration, we can simulate optimal, high-probability safety property development trajectories.

How: Implement a constrained Gaussian Process surrogate structured around a causal SCM. We explicitly define the action node as the zoning intensity policy (A), the outcome as realized housing units (Y), the protest hazard as petition occurrence (P), and index unobserved confounders as U, distinguishing them from the environment or regime $e \in \mathcal{E}$. The constrained causal objective under regime e is:

$$\max_{a \in \mathcal{A}} \mathbb{E}[Y \mid do(A = a), U, e] \quad \text{s.t.} \quad \Pr(P = 1 \mid do(A = a), U, e) \leq \delta$$

Mathematical Description & Strategy

We form a sequential decision-making pipeline where the action a_t maximizes yield without violating the defined safety margin for formalized protests. To ensure mathematical consistency for probabilistic constraints, we define a real-valued latent safety margin $g(a)$ instead of directly assigning a GP to a bounded probability. The surrogate models are Gaussian Processes:

$$\begin{aligned} f(a) &\sim \mathcal{GP}(m_f(a), k_f(a, a')) \quad \text{(Housing Yield Objective)} \\ g(a) &\sim \mathcal{GP}(m_g(a), k_g(a, a')) \quad \text{(Latent Protest Safety Margin)} \end{aligned}$$

We map the latent margin to a protest probability via a link function $\pi(a) = \sigma(g(a))$, enforcing feasibility as $g(a) \leq 0$. Denoting the current best feasible observation (incumbent) as f^*, the Expected Improvement (EI) evaluates the expected gain:

$$\alpha_{\text{EI}}(a) = \mathbb{E}[\max(0,\ f(a) - f^*) \mid \mathcal{D}_t]$$

which has a closed form driven by the GP posterior mean and variance (**?**). Assuming independence between the housing yield objective and the protest safety margin, Constrained EI factorizes multiplicatively: $\alpha_{\text{CEI}}(a) = \alpha_{\text{EI}}(a) \cdot \Pr(g(a) \leq 0)$. However, in zoning regimes, housing yield and protest risk are fundamentally correlated. If independence fails, joint modeling of objective and constraints (e.g., multi-task GPs) is necessary to compute true expected feasible improvement. Furthermore, because zoning approvals are noisy and outcomes are delayed, we must employ algorithms that explicitly address noisy EI and noisy constraints (e.g., Safe Exploration frameworks like SAFEOPT

(?)), providing high-probability safety guarantees rather than evaluating deterministic simulator bounds.

Integration with Current SCM Framing

This formulation enhances the SCM structures within my thesis draft (????). Rather than a flat concept shift assumption, we formally model the 2022 Austin municipal election turnover as a structural change point altering the causal graph's parameters under Dynamic Causal BO (DCBO) (?). Instead of continuous drift, this change-point formalization allows us to analyze how the discrete policy handover structurally modified the mechanisms determining protest hazards, resetting the surrogate belief over the interventional estimand.

Next Steps

I will begin mapping the 7,074 historical discretionary zoning cases into a simulated sequential optimization pathway, constructing a post-hoc analysis to test whether the historical sequence of council approvals was systematically more exploratory or exploitative than an idealized safe BO algorithm (?).

References